

POSTERIOR UVEAL MELANOMA IN YOUNG PATIENTS TREATED WITH PROTON BEAM THERAPY

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Purpose: The purpose was to study the clinical profile and prognosis of young patients with uveal melanoma treated by proton beam irradiation.

Methods: A retrospective case-control series was studied. Seventeen patients aged ≤ 20 years with uveal melanoma and 51 risk factor-matched control subjects aged > 20 years were treated for posterior uveal melanoma between 1975 and 2004. All patients were treated with proton beam irradiation at Harvard Cyclotron. Metastasis and survival of patients were the main outcome measures.

Results: Of a total of 3,207 patients with uveal melanoma, 17 patients were ≤ 20 years of age at the time of diagnosis. No metastatic deaths were observed (median follow-up, 16 years). Among the control group, there were 7 of 50 metastasis-related deaths.

Conclusion: There was an excellent outcome regarding metastasis in a cohort of young patients with posterior uveal melanoma treated with proton beam irradiation.

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Uveal melanoma is the most common primary intraocular tumor in adults. The reported incidence in the United States is between 4.8 and 10.9 per million representing 2.9% of all recorded cases of melanoma.¹ Mean age at diagnosis is 55 years, and it is rare in children and teenagers accounting for $\sim 1\%$ of all cases.^{2–11} Congenital uveal melanoma is very infrequent with only four well-documented cases reported in the literature.^{12–15}

Previous studies of young patients with uveal melanoma have suggested improved survival compared with adults.^{4,7–10,16} With the exception of the study by Shields et al⁹ (96% survival rate), in which only 12% of the tumors were located in the iris, the rest had a significant proportion (20–50%) of iris tumors, which have a favorable prognosis compared with ciliary body and choroidal tumors. Plaque irradiation and enucleation were the most frequent treatments in these studies.

This study describes the clinical characteristics and survival in young patients (≤ 20 years) with choroidal and ciliochoroidal melanoma treated with proton beam irradiation at the Harvard Cyclotron and Francis H. Burr Proton Therapy Center at Massachusetts General Hospital between July 1975 and December 2004.

Methods

This was a retrospective case-control series. Charts of all patients treated with proton beam irradiation for uveal melanoma between July 1975 and December 2004 were reviewed. Diagnosis was based on ophthalmoscopic, fluorescein angiographic, and ultrasonographic appearance. No biopsy was performed. Patients aged ≤ 20 years with melanoma of the ciliary body and/or choroid were included in this analysis. The patient data included age, race, sex, and the eye involved. Information regarding presence of symptoms, systemic disease, metastatic disease, and personal or family history of neoplasm in first-degree relatives was recorded. Tumor location was categorized according to anterior extent: tumors were coded as being located in the posterior pole, at the equator, between the equator and ciliary body, and involving the ciliary body. Tumor size was determined by measuring the maximum basal diameter and the

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Table 1. Clinical Characteristics of Patients Aged ≤ 20 Years With Uveal Melanoma Receiving Proton Beam Irradiation

Case No.	Age (Years)/Sex	Tumor Location	Symptoms at Presentation	Tumor LTD (mm)	Tumor Height (mm)	Follow-Up (Years)	Status/Final VA
1	14/M	Ch	Y	9	5.1	25.2	A/R/HM
2	16/F	CB	N	10	5.8	19.6	A/R/20/40
3	17/M	Ch	Y	11	4.6	21.3	A/R/20/320
4	18/M	Ch	Y	11	3.8	22.7	A/R/20/800
5	20/F	Ch	Y	11	3.6	21.4	A/R/LP
6	19/M	Ch	Y	11	2.6	15.9	A/R/HM
7	18/M	Ch	Y	12	8	9.9	A/R/20/50
8	20/F	Ch	Y	12	8	24	A/R/CF
9	14/F	Ch	N	13	9	12	A/R/NR
10	20/F	Ch	Y	14	2	24.5	A/R/NR
11	11/F	Ch	Y	16	7.6	5.2	A/R/20/80
12	19/F	Ch	Y	16	6	16	A/R/20/60
13	19/F	Ch	N	16	3	4.7	A/R/NR
14	20/M	Ch	Y	16	4.7	6.8	A/R/20/25
15	16/M	Ch	Y	16	10.5	10.8	A/R/CF
16	15/F	Ch	Y	16	10	16.3	A/R/NLP
17	15/F	Ch	N	17	9	23.1	A/R/NR
Median	18	—	76%yes	13	5.8	16	—

LTD, largest tumor diameter; VA, visual acuity; M, male; F, female; Ch, choroid; CB, ciliary body; Y, yes; N, no; A, alive; R, eye retained; HM, hand motion; LP, light perception; CF, count fingers; NR, not recorded; NLP, no light perception.

maximum tumor thickness assessed by clinical and ultrasound examination. All patients were treated with proton beam irradiation. Screening for metastatic disease involved liver function tests annually. Abnormal values were followed with liver imaging. The most recent follow-up information on the patient's vital status (alive, alive with metastases, dead from metastases, and dead from other causes) was recorded. Survival status was ascertained by active surveillance of patients or a search of the Social Security Death Index. Outcomes were compared with a group of control subjects frequency-matched (3:1) on the following risk factors for mortality: largest tumor diameter, pigmentation of tumor, presence of symptoms, eye color, and tumor location.

Results

Among 3,207 patients with uveal melanoma treated between July 1975 and December 2004, there were 17 patients (0.53%) aged ≤ 20 years. Median age at diagnosis was 18 years (range, 11–20 years). Sixteen patients (94%) were white, and 1 patient (6%) was Hispanic. There were 7 male and 10 female patients. All tumors were unilateral with 16 involving only the choroid and 1 involving the choroid and ciliary body. There were no iris melanomas. No other predisposing conditions such as oculodermal melanocytosis or dysplastic nevus syndrome were noted. No patients had a history of another malignancy. Patients were observed for 5 years to 25 years (median, 16 years). Tables 1 and 2 summarize the clinical data of these patients.

Our control group of older patients ($n = 51$) was matched on largest tumor diameter, degree of pigmentation, presence of symptoms, eye color, and tumor location. Their median age was 57 years (range, 26–81 years). They were followed for 3 years to 25 years (median, 16.7 years). There were 7 of 50 deaths related to melanoma in the control group (unknown cause for 1 patient) and none in the cohort of young patients. Kaplan-Meier survival curves are shown in Figure 1. One third of the young patients saw an initial improvement in visual acuity postoperatively. Two thirds of the young patients experienced vision loss to levels $< 20/200$.

Discussion

Several reports of malignant uveal melanoma in children and adolescents have been published in the past, suggesting that survival is better in younger patients (Table 3). In most of these reports, a high proportion of iris melanoma cases, which have a more favorable prognosis, were included in the analysis. Our study confirms that younger patients with melanomas of the ciliary body/choroid also have excellent long-term outcomes with 100% survival over a median follow-up of 16 years. Unfortunately, however, the majority does not attain good visual acuity.

In our series, the ratio of melanoma in patients aged ≤ 20 years was 0.53%, which is slightly lower than that observed in previous reports (0.79–1.44%; Table 3). This could reflect a slight referral bias as a result of the perception that children may be poor candidates for

Table 2. Patient and Tumor Characteristics of Young Patients With Ocular Melanoma and an Older Control Group

	Young Patients	Risk Factor-Matched Older Age Control Group
Median age at diagnosis in years (range)	18 (11–20)	57 (26–81)
Race, no. (%)		
White	16	46
Hispanic	1	1
Eye color, no. (%)*		
Dark	6 (40)	19 (40)
Medium	3 (20)	8 (17)
Light	6 (40)	21 (43)
Median height in mm (range)	5.8 (2.6–10.6)	5.4 (1.9–12.2)
Median LTD in mm (range)*	13 (9.0–17.0)	13 (8.0–17.0)
Tumor pigmentation, no. (%)*		
Heavy	8 (57)	24 (57)
Moderate	3 (21)	9 (21)
Minimal	3 (21)	9 (21)
Tumor origin, no. (%)*		
Ciliary body	1 (6)	3 (6)
Choroidal	16 (94)	48 (94)
Symptoms at presentation, no. (%)*		
Patients with other primary cancers, no. (%)	0	5 (10)
Follow-up years, median (range)*	16 (4.7–25.2)	16.7 (2.7–24.5)
No. patients with cancer in another family member (%)	11 (65)	35 (69)

*Risk factors on which cases and control subjects were matched. LTD, largest tumor diameter.

proton beam irradiation because of the necessity for fixation during treatments. However, we demonstrate that patients aged ≤ 11 years have been successfully treated with this modality.

Because early onset of malignancy can be a feature of genetic predisposition, it is useful to investigate the occurrence of other associated conditions in children/teenagers with uveal melanoma. Although we found no predisposing conditions in our patients, others have reported an increased prevalence of oculodermal melanocytosis among younger patients with melanoma compared with adults. Seven of 63 (11%) in the report by Singh et al¹⁰ and 4 of 17 in Verdaguer's series¹¹ had oculodermal melanocytosis. This is severalfold higher than the prevalence of $\sim 1.4\%$ in the adult population of patients with melanoma.¹⁷ It has been proposed that an

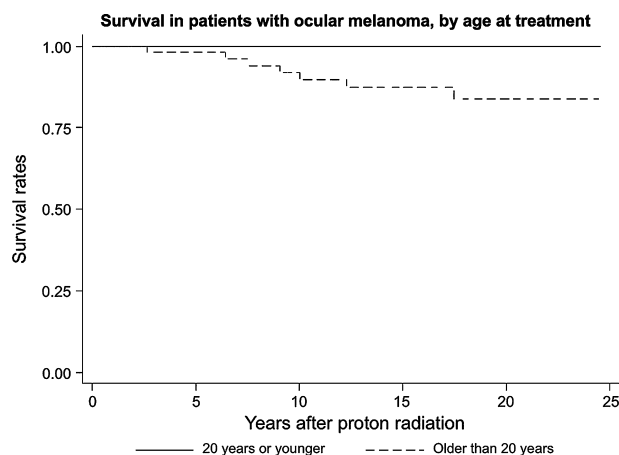


Fig. 1. Kaplan-Meier survival estimates of 17 young patients with uveal melanoma and 51 risk factor-matched older patients with uveal melanoma.

overall increased number of melanocytes may predispose these patients to malignancy. The lifetime risk of developing uveal melanoma in patients with oculodermal melanocytosis is 1 in 400 compared with 1 in 13,000 for the general population.¹⁸ However, among these patients, the median age of diagnosis is similar to that of a randomly selected uveal melanoma patient population, and only 0.5% of patients with oculodermal melanocytosis develop uveal melanoma before age 10 years.¹⁸

Familial atypical mole and melanoma (FAMM) syndrome, also known as dysplastic nevus syndrome, is an autosomal-dominant condition in which almost 100% of the patients develop cutaneous melanoma in their lifetime. An association between uveal melanoma and FAMM has been suggested but remains unconfirmed. Greene et al¹⁹ examined 26 adult patients with FAMM and did not find either uveal melanoma or suspicious choroidal nevi. Taylor et al²⁰ examined 44 adult patients with uveal melanoma and found no increased incidence of dysplastic nevi compared with the general population. However, Singh et al²¹ found that their 8 patients with FAMM and uveal melanoma (of the total 4,600 patients with uveal melanoma) had an average age of diagnosis ~ 15 years younger than the overall group of patients with melanoma (age, 40 vs. 55 years). The younger age of presentation was consistent with their review of the seven other similar cases reported in the literature, suggesting a genetic predisposition for the development of melanoma. Two of 63 patients in the pediatric series of Singh et al¹⁰ and no patients in our current series had FAMM syndrome, suggesting that it is not a major risk factor for uveal melanoma in the pediatric population. However, our data are limited by the retrospective nature of our study and the fact that the data about FAMM were collected by patients' answers to a written questionnaire.

Table 3. Summary of Cases of Pediatric and Adolescent Uveal Melanoma in the Literature

References	Iris Cases/ Deaths (% Mortality)	CB-Ch Cases/ Deaths (% Mortality)	Treatment (Number When Known)	Ratio (Younger Cases/All Cases of Uveal Melanoma; %)	Follow-Up (Years)
Verdaguer ^{11*}	30/1 (3.3)	38/8 (21.1)	Enucleation/excision	46/3,628† (1.27)	?
Verdaguer ^{11‡}	6/0 (0)	11/0 (0)	Enucleation/excision	?§	5
Barr et al ³	27/4 (14.8)	42/13 (31)	Enucleation/excision	92/6,358¶ (1.45)	16
Gailloud et al ⁴	3/0 (0)	18/0 (0)	Excision (4), proton RT(11), Co plaque (3)	21/1,450 (1.45)	2.8
Singh et al ¹⁰	16/0 (0)	47/4 (8.5)	Enucleation/excision (51), RT (9)	63/8,000 (0.79)	4.25
Pogrzebielski et al ⁷	6/0 (0)	5/0 (0)	Enucleation (2), RT (9)	11/800 (1.38)	5.1
Present study	0/0 (N/A)	17/0 (0)	Proton RT	17/3,207 (0.53)	16
All without Verdaguer Row 1	58/4 (6.9)	140/17 (12)	—	204/19,815 (1)	
All	88/5 (5.7)	178/25 (14)	—	250/23,443 (1.1)	

*1935–1965 literature reviewed by Verdaguer (includes Apt L. AFIP 1962).

†Only from Apt L.

‡A total of 1965 cases from the New York Institute of Ophthalmology.

§Could not be calculated.

¶Prognosis was reported in only 69 cases because of lack of follow-up in the remainder of cases.

||Excluding the review of literature by Verdaguer¹¹ (row 1 above) most of which was also included in Barr et al.³

CB-Ch, ciliochoroidal; N/A, not applicable; RT, radiotherapy; Co plaque, plaque brachytherapy with cobalt 60.

It is interesting to note that at least four independent studies^{4,7,10,11} and our data suggest that the younger patients have better prognoses. Younger patients have a more robust immune system and better immune surveillance that may keep the tumor under control. It is also possible that there is something intrinsic to tumors that arise early in life that makes these tumors more sensitive to treatment. Adult patients with cutaneous melanoma have a more favorable prognosis the younger they are.²² Younger patients do tend to have a more favorable clinicopathologic presentation²³ along with better lymphatic function.²⁴ Pediatric cutaneous melanoma analysis also shows better prognoses with younger age with a 91% 5-year survival rate.²⁵

In summary, uveal melanoma rarely occurs in children/teenagers (0.5–1.5% of all uveal melanomas). The absence of multiple features of genetic predisposition such as bilaterality, multifocality, multiplicity of primary tumors, and family history suggest that inheritable genetic traits play a minimal role in the pathogenesis of uveal melanoma in this younger population. When this tumor does affect pediatric patients, the prognosis for survival is excellent with high rates of long-term survival achieved after globe-sparing therapy. Therefore, younger age should not preclude radiation therapy for uveal melanoma.

Key words: uveal, melanoma, young, proton beam irradiation.

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